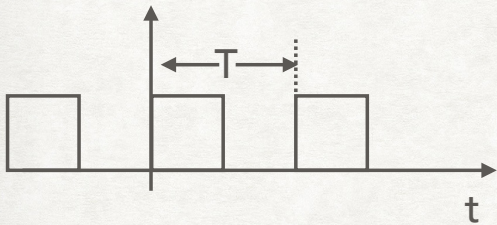


CONTINUOUS-TIME FOURIER TRANSFORM

- Describe both periodic & non-periodic signals and tools for analysis of LTI system
- Represent aperiodic signals, such as $u(t)$, $\exp[-t]u(t)$, $\text{rect}(t/T)$ $\rightarrow$ decompose signals into a set of weighted sum of complex exponentials.
- Aperiodic signal can be viewed as a periodic signal, with an infinite period. In the Fourier series representation of a periodic signal, as the period increases, the fundamental frequency decreases and the harmonically related components become closer in frequency. As the period becomes infinite, the frequency components form a continuum and the Fourier series sum becomes integral.

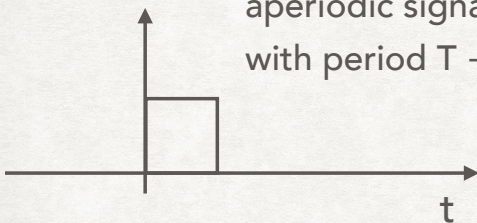
CONTINUOUS-TIME FOURIER TRANSFORM

$\tilde{x}(t)$ periodic signal with period T



$x(t)$

aperiodic signal $\rightarrow$ periodic signal
with period $T \rightarrow \infty$



Considering the periodic signal $\tilde{x}(t)$, increase $T \rightarrow \infty$, only a single pulse remains.

$$\text{Fourier representation of } \tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t} \quad (3.14)$$

$$\text{where } a_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt \quad (3.15)$$

In the limit as $T \rightarrow \infty$, we see that $\omega_0 = 2\pi/T$ becomes infinitesimally small quantity $d\omega$, so that

$$\frac{1}{T} \rightarrow \frac{d\omega}{2\pi} \quad (3.16)$$

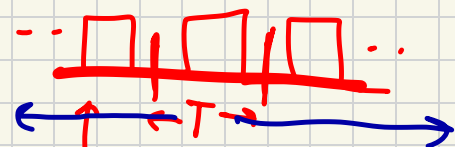
and in the limit in (3.15) $n\omega_0$ should be continuous variable $\rightarrow \omega$, Fourier coefficients per unit frequency interval are

$$a_n = \frac{d\omega}{2\pi} \int_{t=-\infty}^{\infty} \tilde{x}(t) e^{-j\omega t} dt \quad (3.17)$$

$$\tilde{x}(t)$$

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T} \int_{\langle T \rangle} \tilde{x}(t) e^{-jk\omega_0 t} dt$$



$$\Delta\omega \rightarrow \underline{\underline{d\omega}}$$

$$\tilde{x}(t) \approx \tilde{x}(t) \text{ for } T \text{ mu}$$

$$T \rightarrow \infty$$

$$\omega_0 = \frac{2\pi}{T} \rightarrow \frac{1}{T} \varepsilon \frac{\omega_0}{2\pi}$$

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} \left[\frac{1}{T} \int_{\langle T \rangle} \tilde{x}(t) e^{-jk\omega_0 t} dt \right] \cdot e^{jk\omega_0 t}$$

fairly mu

$$x(t) = \int_{\omega=-\infty}^{\infty} \left[\frac{d\omega}{2\pi} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right] \cdot e^{j\omega t}$$

$$k \rightarrow \underline{\underline{\omega}}$$

$$k = -\infty \rightarrow \omega = k\omega_0 = -\infty$$

$$k = \infty \rightarrow \omega = \infty$$

$$x(t) = \frac{1}{2\pi} \int_{\omega=-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t} d\omega$$

$$X(j\omega)$$

$$X(j\omega) \leftarrow \int_{t=-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) \leftrightarrow X(j\omega)$$

peri

$$x(t) \leftrightarrow a_n$$

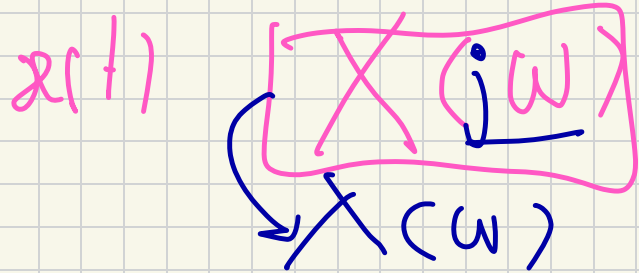
$$x(t) \leftrightarrow X(j\omega)$$

aper.

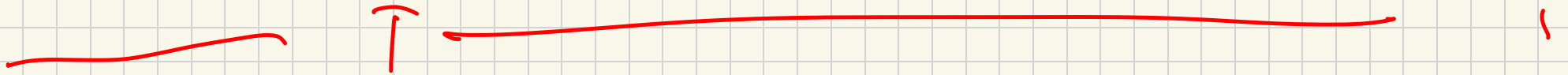
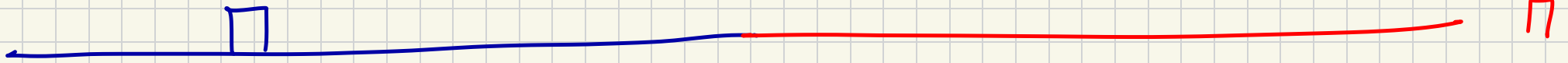
(2n)

FT

$$\text{FT } x(t) \rightarrow \{x(t)\} = X(j\omega)$$



$\tilde{x}(t)$



$x(t)$

$\square$



CONTINUOUS-TIME FOURIER TRANSFORM

Substituting (3.17) into (3.14), the limit of the sum becomes an integral, and $\tilde{x}(t)$ approaches $x(t)$, we obtain

$$x(t) = \int_{\omega=-\infty}^{\infty} \boxed{\left[\int_{t=-\infty}^{\infty} x(t) e^{-j\omega t} dt \right]} e^{j\omega t} \frac{d\omega}{2\pi} \quad (3.18)$$

The inner integral in the blankets is a function of ω only, not t . Denoting this integral by $X(j\omega)$ and we can write (3.18) as

Fourier-transform pair for aperiodic signal:

8) as

m pair

signal:

and

$$\underline{x(t)} = \frac{1}{2\pi} \int_{\omega=-\infty}^{\infty} \underline{X(j\omega)} e^{j\omega t} d\omega$$
$$\left[\begin{array}{l} X(j\omega) = \int_{t=-\infty}^{\infty} \underline{x(t)} e^{-j\omega t} dt \end{array} \right.$$

$\omega \sim \omega$

ω

CT

$\underline{x(t)} \leftrightarrow \underline{a_n}$

$\underline{x(t)} \leftrightarrow \underline{X(j\omega)}$

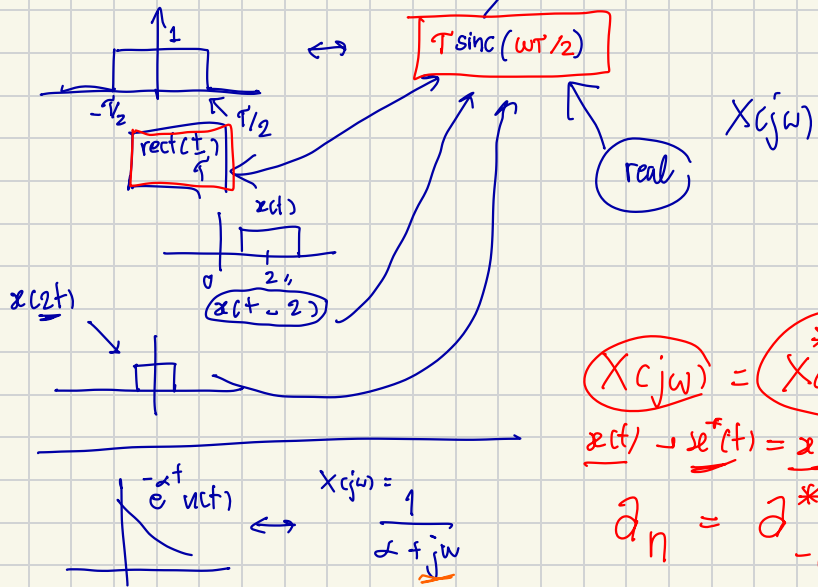
Cont fr

(3.19)

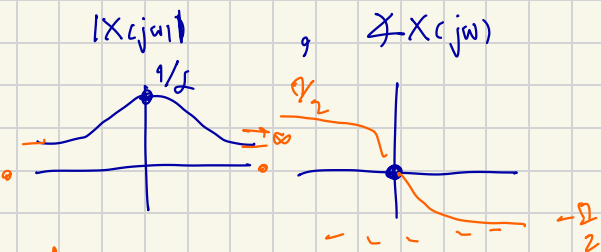
(3.20)

$$X(j\omega) = \int_{t=-\infty}^{\infty} \underline{x(t)} e^{-j\omega t} dt \quad (3.20)$$

time $\leftrightarrow$ freq. $\sin x = \frac{\sin x}{x}$



$$\begin{aligned} \underline{x(t)} &\rightarrow \underline{x^*(t)} = \underline{x(t)} \\ a_n &= a_{-n}^* \end{aligned}$$



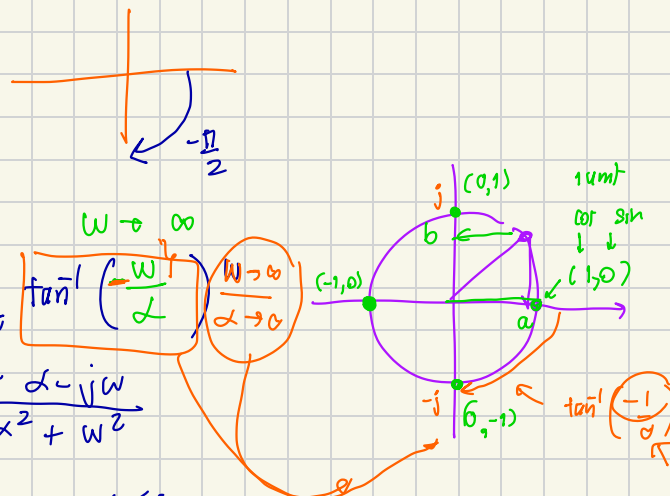
$$W \rightarrow \infty$$

$\left| \frac{1}{s + j\infty} \right| = 0 \quad \checkmark$
 $\frac{1}{j\infty} \rightarrow 0$

$\tan^{-1}(\quad)$

$$X_{Cj\omega} = \frac{1}{\omega + j\omega} \cdot \frac{\omega - j\omega}{\omega - j\omega} =$$

$\lim_{w \rightarrow \infty} \tan^{-1} \left(\frac{w}{L} \right) = \frac{\pi}{2}$
 $\frac{L - jw}{L^2 + w^2}$



$$z = a + bj \quad |z| = \sqrt{a^2 + b^2} \quad \angle z = \tan^{-1}\left(\frac{b}{a}\right)$$

$$\begin{aligned} x(t) &\leftrightarrow X(j\omega) \\ \delta(t) &\leftrightarrow 1 \end{aligned}$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

$$X(j\omega) \leftrightarrow X(\omega)$$

CONTINUOUS-TIME FOURIER TRANSFORM

One-to-one correspondence between $x(t)$ and $X(j\omega)$: $x(t) \leftrightarrow X(j\omega)$ also see $x(t) \leftrightarrow X(\omega)$

• $X(j\omega)$ is often denoted by $\mathcal{F}\{x(t)\}$ $\mathcal{F}\{x(t)\} = X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$

• One of condition sets, referred to as the Dirichlet conditions to guarantee that a signal has Fourier transform, require that

- * (1) $x(t)$ is absolutely integrable; that is, $\int_{-\infty}^{\infty} |x(t)| dt < \infty$
- (2) $x(t)$ has a finite number of minima and maxima within any finite interval
- (3) $x(t)$ has a finite number of discontinuities within any finite interval. Furthermore, each of this interval must be finite.

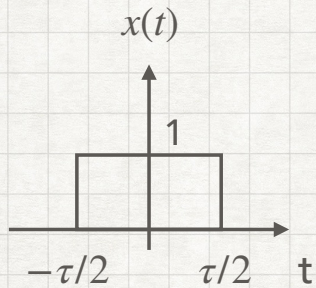
However, there are more signals that are not integrable (periodic signal, unit step), but they also have Fourier transform

FOURIER TRANSFORM

EXERCISE #3.4



Find continuous-time Fourier transform of $x(t) = \text{rect}(t/\tau)$

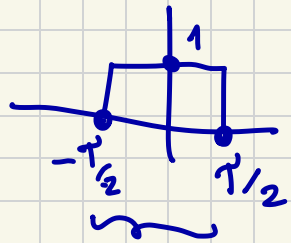


From

$$\begin{aligned} X(j\omega) &= \int_{t=-\infty}^{\infty} x(t)e^{-j\omega t} dt \\ &= \int_{t=-\tau/2}^{\tau/2} (1)e^{-j\omega t} dt \\ &= \left. \frac{e^{-j\omega t}}{-j\omega} \right|_{t=-\tau/2}^{\tau/2} \\ &= \frac{e^{-j\omega(\tau/2)} - e^{-j\omega(-\tau/2)}}{-j\omega} \\ &= \frac{2 \sin(\omega\tau/2)}{\omega} = \frac{\tau \sin(\omega\tau/2)}{(\omega\tau/2)} \\ &= \tau \text{sinc}(\omega\tau/2) \end{aligned}$$

$$\text{From } \text{sinc}(x) = \frac{\sin(x)}{x}$$

$$x(t) = \text{rect}(t/\tau)$$



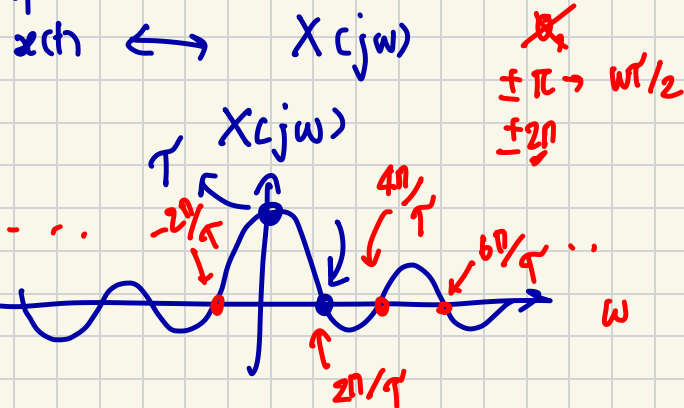
$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\ &= \int_{-\tau/2}^{\tau/2} 1 \cdot e^{-j\omega t} dt \\ &= \left. \frac{e^{-j\omega t}}{-j\omega} \right|_{-\tau/2}^{\tau/2} \\ &= \frac{1}{-j\omega} \left(e^{-j\omega(\tau/2)} - e^{+j\omega(\tau/2)} \right) \\ &= \frac{1}{-j\omega} \left(-e^{-j\omega(\tau/2)} + e^{+j\omega(\tau/2)} \right) \\ &= \frac{2 \sin(\omega\tau/2)}{\omega} \\ &= \tau \cdot \text{sinc}(\omega\tau/2) \\ &= \tau \text{sinc}(\omega\tau/2) \end{aligned}$$

$$\frac{e^{jx} - e^{-jx}}{2j} = \sin x$$

$$\text{sinc}(x) = \frac{\sin x}{x}$$

$$\text{sinc}(x) = \frac{\sin x}{x \cdot (\tau/2)}$$

$$\text{rect}\left(\frac{t}{\tau}\right) \longleftrightarrow \tau \text{sinc}(\omega\tau/2) \quad \omega = \frac{2\pi}{\tau}$$



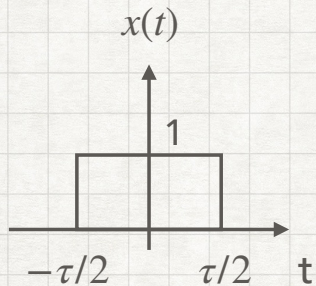
$$\begin{aligned} \text{Re}\{X(j\omega)\} & \\ \text{Im}\{X(j\omega)\} & \\ |X(j\omega)| & \\ \angle X(j\omega) & \end{aligned}$$

FOURIER TRANSFORM

EXERCISE #3.4



Find continuous-time Fourier transform of $x(t) = \text{rect}(t/\tau)$



From

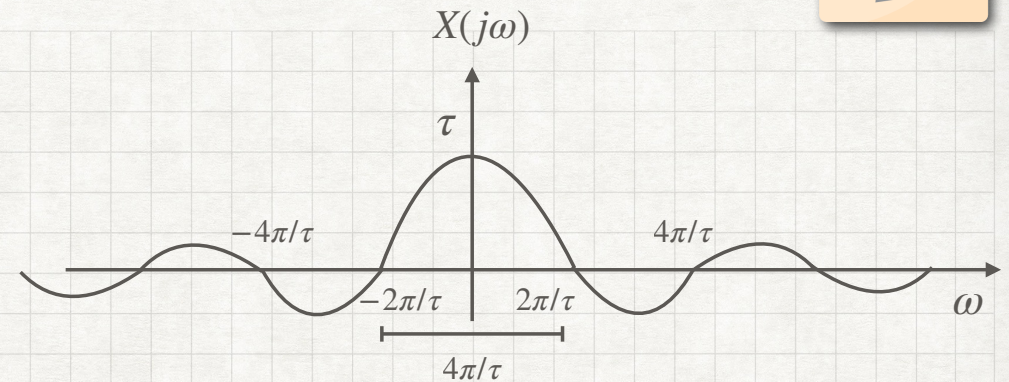
$$X(j\omega) = \int_{t=-\infty}^{\infty} x(t)e^{-j\omega t} dt$$
$$= \int_{t=-\tau/2}^{\tau/2} (1)e^{-j\omega t} dt$$

$$= \left. \frac{e^{-j\omega t}}{-j\omega} \right|_{t=-\tau/2}^{\tau/2}$$
$$= \frac{e^{-j\omega(\tau/2)} - e^{-j\omega(-\tau/2)}}{-j\omega}$$

$$= \frac{2 \sin(\omega\tau/2)}{\omega} = \frac{\tau \sin(\omega\tau/2)}{(\omega\tau/2)}$$

From $\text{sinc}(x) = \frac{\sin(x)}{x}$

$$= \tau \text{sinc}(\omega\tau/2)$$



FOURIER SERIES VS. FOURIER TRANSFORM

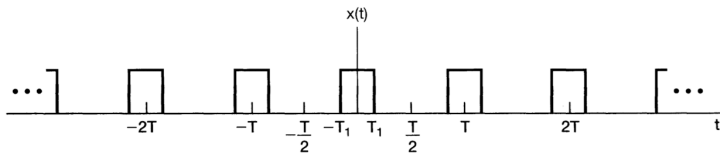
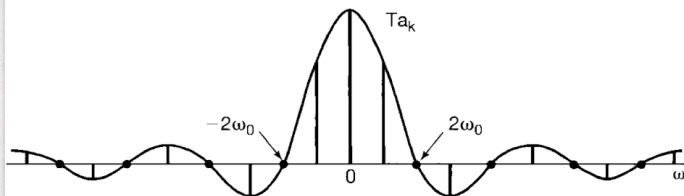
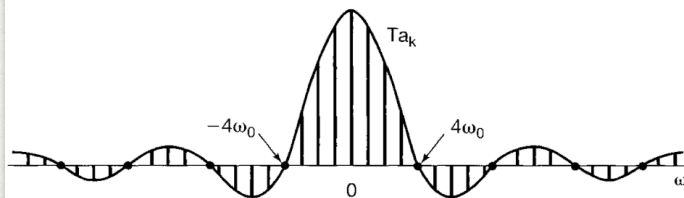


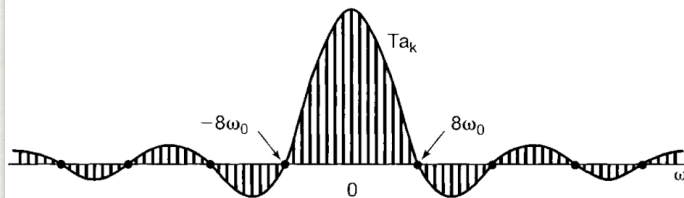
Figure 4.1 A continuous-time periodic square wave.



(a)

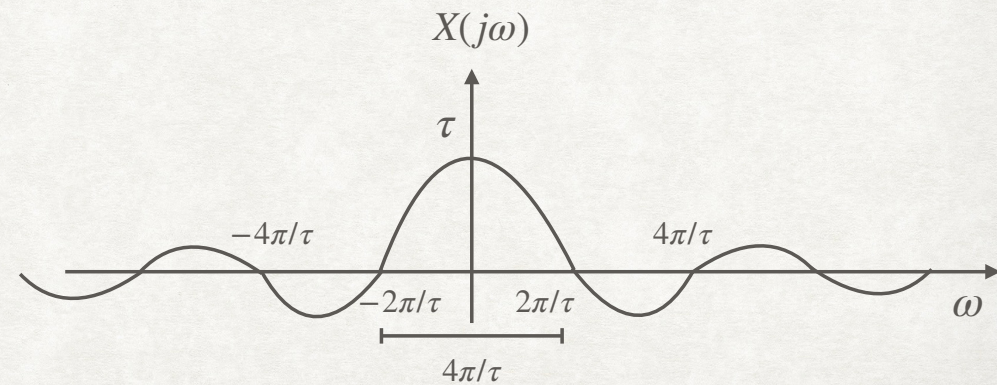
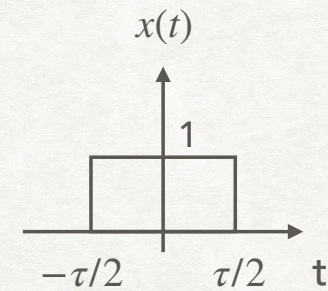


(b)



(c)

Figure 4.2 The Fourier series coefficients and their envelope for the periodic square wave in Figure 4.1 for several values of T (with T_1 fixed): (a) $T = 4T_1$; (b) $T = 8T_1$; (c) $T = 16T_1$.

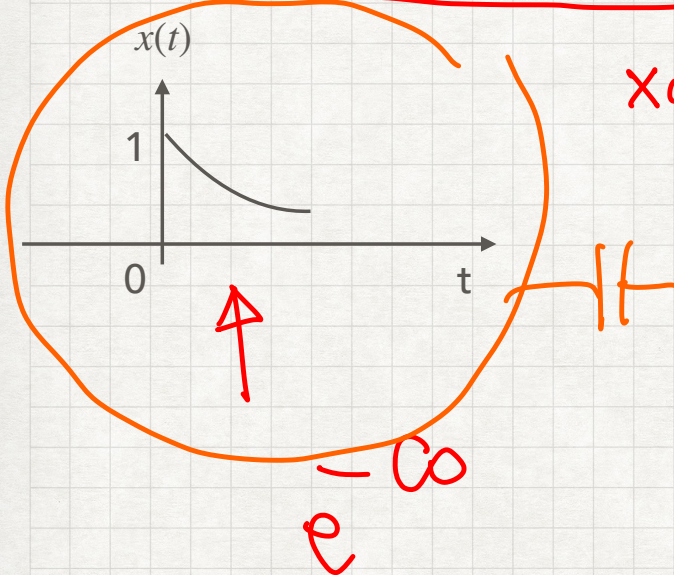


FOURIER TRANSFORM

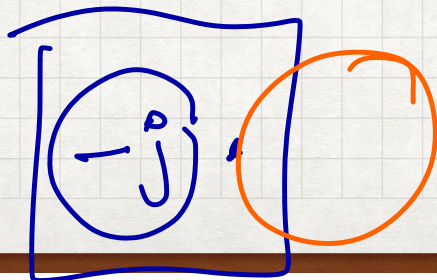
EXERCISE #3.5



Find continuous-time Fourier transform of $x(t) = \exp[-\alpha t]u(t)$ where $\alpha > 0$

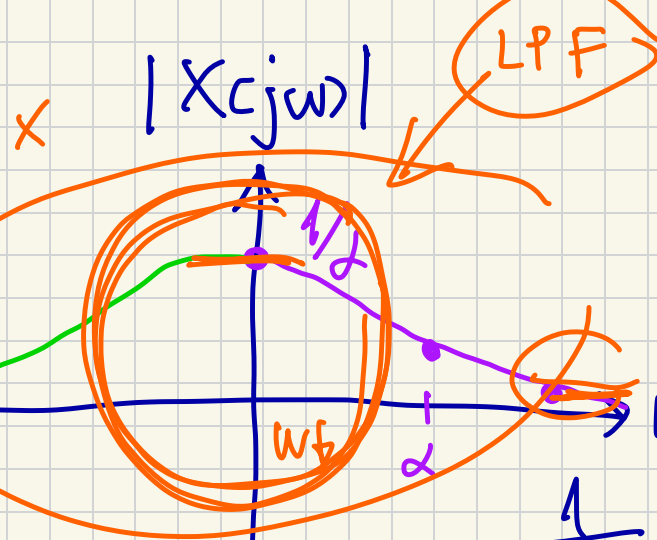


$$\begin{aligned}
 X(j\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\
 &= \int_0^{\infty} e^{-\alpha t} e^{-j\omega t} dt \\
 &= \left. \frac{e^{-t(\alpha + j\omega)}}{-(\alpha + j\omega)} \right|_0^{\infty} \\
 &= 0 + \boxed{\frac{1}{\alpha + j\omega}} \quad \star
 \end{aligned}$$

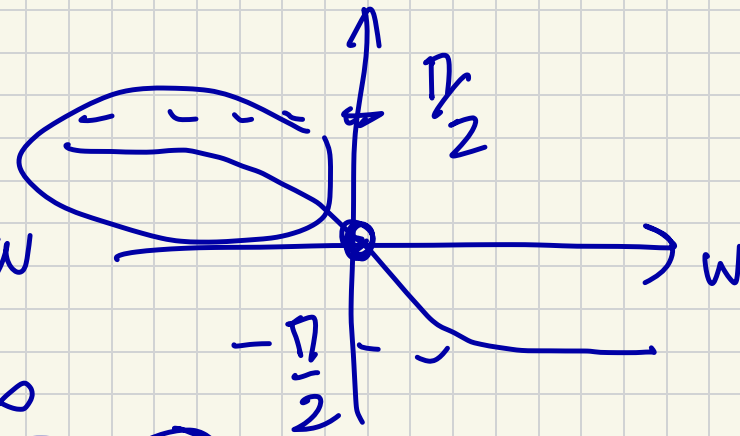


$$\frac{1}{\alpha + j\omega} = \frac{\alpha - j\omega}{(\alpha - j\omega)(\alpha + j\omega)} = \frac{\alpha - j\omega}{\alpha^2 + \omega^2}$$

$$X_c(j\omega) = \frac{1}{\alpha + j\omega}$$



$$\angle X_c(j\omega)$$



$|X_c(j\omega)|$
specch $\rightarrow \omega$

$$\angle X_c(j\omega)$$

$$\omega=0$$

$$X_c(j\omega) = \frac{1}{\alpha}$$

$$\omega \rightarrow \infty$$

$$X_c(j\omega) = \frac{1}{\alpha + j\omega}$$

$$\angle X_c(j\omega) = 0$$

$$\frac{1}{\alpha + j\infty}$$

$$-j\infty$$

$$\frac{1}{j\infty} = -j0$$

$$-\frac{\pi}{2}$$

$$\omega = \alpha \quad |X_c(j\omega)| = \left| \frac{1}{\alpha + j\alpha} \right| = \left| \frac{\alpha - j\alpha}{\alpha^2 + \alpha^2} \right| = \left| \frac{\alpha - j\alpha}{2\alpha^2} \right| = \sqrt{\left(\frac{\alpha}{2\alpha^2}\right)^2 + \left(\frac{\alpha}{2\alpha^2}\right)^2}$$

$$= \frac{1}{\sqrt{2}\alpha}$$

$$X_c(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x^*(t) = x(t)$$

$$X^*_c(j\omega) = \int_{-\infty}^{\infty} (x(t))^* e^{+j\omega t} dt$$

$$X(-j\omega) \rightarrow X(j\omega)$$

$$\omega \rightarrow -\omega$$

$$X^*(-j\omega)$$

$$= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

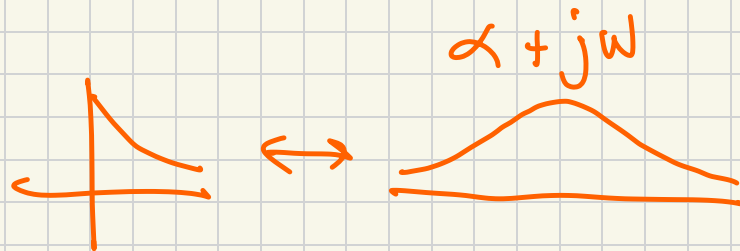
$$|X(j\omega)| = |X(-j\omega)|$$

$$\angle X(j\omega) = -\angle X(-j\omega)$$

$$X(j\omega) = X^*(-j\omega)$$

$$x(t) \leftrightarrow X(j\omega)$$

$$e^{-\alpha t} u(t) \leftrightarrow \frac{1}{\alpha + j\omega}$$



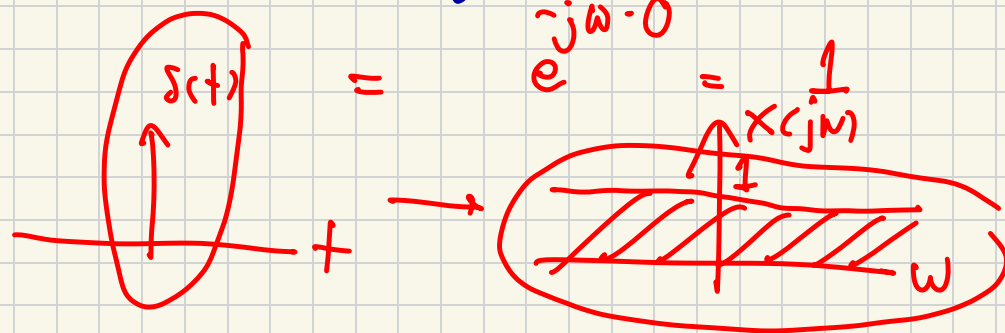
$$X(j\omega) = X^*(-j\omega)$$

$$x(t) \longleftrightarrow X(j\omega)$$

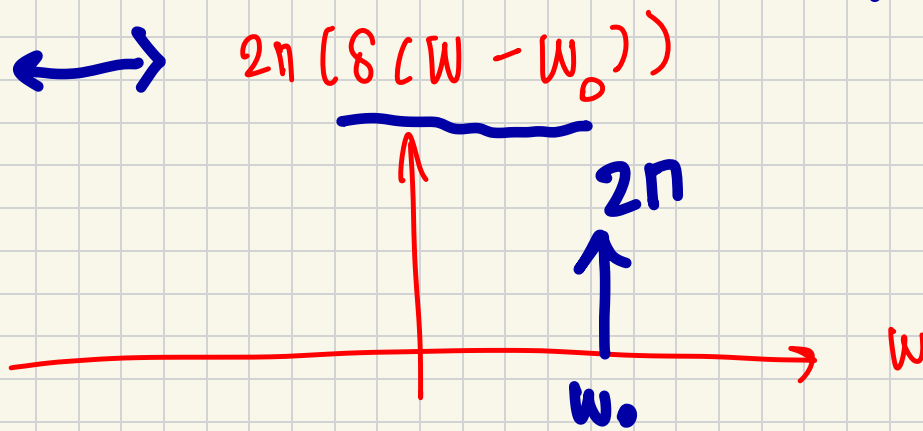
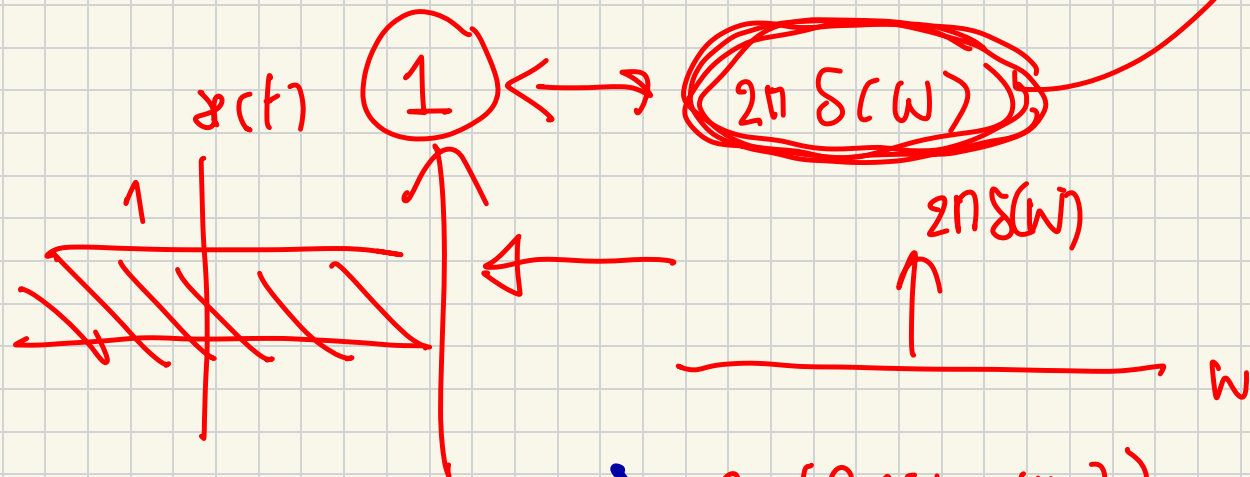
$$s(t) \longleftrightarrow X(j\omega)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\int x(t) \delta(t - t_0) dt = x(t_0)$$



$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(\omega) e^{j\omega t} d\omega$$



$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(\omega - \omega_0) e^{j\omega t} d\omega$$

$$e^{j\omega_0 t} \longleftrightarrow X(j\omega)$$

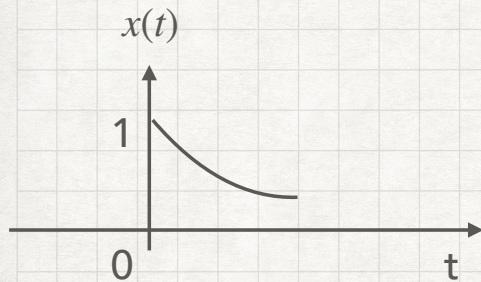
$$e^{j\omega_0 t} \longleftrightarrow 2\pi \delta(\omega - \omega_0)$$

FOURIER TRANSFORM

EXERCISE #3.5



Find continuous-time Fourier transform of $x(t) = \exp[-\alpha t]u(t)$ where $\alpha > 0$



$$\text{From } X(j\omega) = \int_{t=-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-\alpha t} e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{(-\alpha - j\omega)t} dt$$

$$= \left. \frac{e^{(-j\omega - \alpha)t}}{-j\omega - \alpha} \right|_{t=0}^{\infty}$$

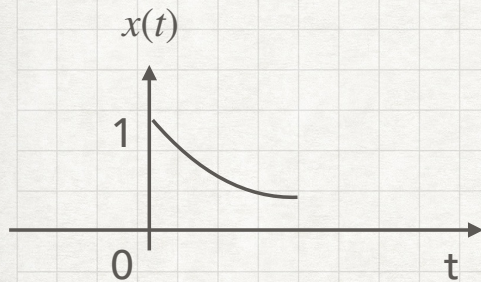
$$= \frac{1}{\alpha + j\omega} \quad \#$$

FOURIER TRANSFORM

EXERCISE #3.5



Find continuous-time Fourier transform of $x(t) = \exp[-\alpha t]u(t)$ where $\alpha > 0$



$$\text{From } X(j\omega) = \int_{t=-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-\alpha t} e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{(-\alpha - j\omega)t} dt$$

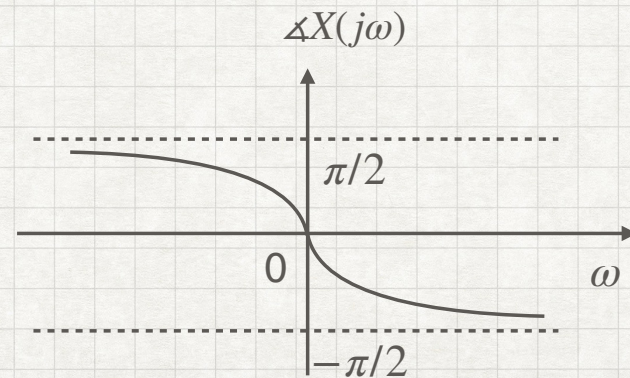
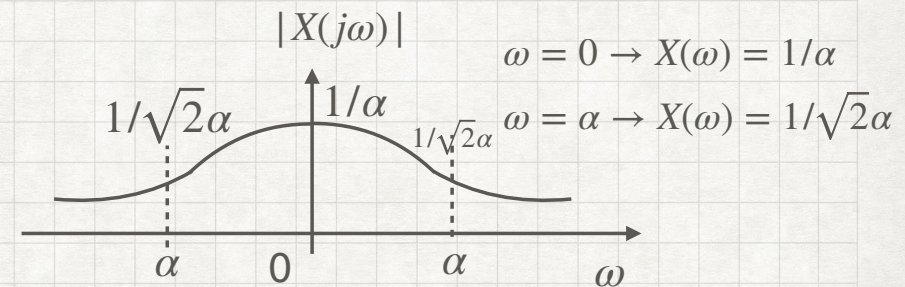
$$= \left. \frac{e^{(-j\omega - \alpha)t}}{-j\omega - \alpha} \right|_{t=0}^{\infty}$$

$$= \frac{1}{\alpha + j\omega} \quad \#$$

$$|X(j\omega)| = \sqrt{(\text{Re}\{X(j\omega)\})^2 + (\text{Im}\{X(j\omega)\})^2}$$

$$\angle X(j\omega) = \tan^{-1}\left(\frac{\text{Im}\{X(j\omega)\}}{\text{Re}\{X(j\omega)\}}\right)$$

We can write $X(j\omega) = |X(j\omega)| e^{j\angle X(j\omega)}$



PROPERTIES OF FOURIER TRANSFORM

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

① *
Linearity

$$ax_1(t) + bx_2(t) \leftrightarrow aX_1(j\omega) + bX_2(j\omega),$$

- Symmetry: if $x(t)$ is a real-valued time signal, then

$$X(-j\omega) = X^*(j\omega)$$

* denotes the complex conjugate

- Time-Shifting & Frequency-Shifting

(Time Shift) $x(t - t_0) \leftrightarrow X(j\omega) e^{-j\omega t_0}$

(Freq. Shift) $x(t) e^{j\omega_0 t} \leftrightarrow X(j(\omega - \omega_0))$

$$\text{rect}\left(\frac{t}{\tau}\right) + e^{-\frac{t}{\tau}} u(t)$$

where a and b are arbitrary constants

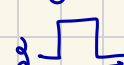
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\omega) e^{j\omega t} d\omega$$

$$X_c(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t - t_0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\omega) e^{j\omega(t - t_0)} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\omega) e^{-j\omega t_0} e^{j\omega t} d\omega$$


$\delta(t) \leftrightarrow X_c(j(\omega - \omega_0))$
 $\text{sinc}(\omega - 2)$



$$X_c'(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$X_c(j(\omega - \omega_0)) = \int_{-\infty}^{\infty} x(t) e^{-j(\omega - \omega_0)t} dt$$

$$= \int_{-\infty}^{\infty} x(t) e^{+j\omega_0 t} e^{-j\omega t} dt$$

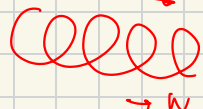
$$x(t) e^{j\omega_0 t} \leftrightarrow X_c'(j(\omega - \omega_0))$$

$$\frac{1}{2} \text{rect}\left(\frac{t}{2}\right) \leftrightarrow \text{sinc}(\omega - 2)$$

$$\text{rect}\left(\frac{t}{T}\right) \leftrightarrow T \text{sinc}(T(\omega - \omega_0))$$

$$\frac{1}{2} \text{rect}\left(\frac{t}{2}\right) \cdot e^{j\omega_0 t} \leftrightarrow \text{sinc}(\omega - 2) \quad T=2$$

$$e^{-j\omega t_0} = \cos + j \sin$$



$$x(t) \leftrightarrow X_c(j\omega)$$

$$x(t - t_0) \leftrightarrow X_c(j\omega) \cdot e^{-j\omega t_0}$$

$$x(t) = \text{rect}\left(\frac{t}{T}\right) \leftrightarrow X_c(j\omega) = T \text{sinc}(\omega T/2)$$

$$\text{rect}(t - 2) \leftrightarrow ?$$

$$x(t - t_0) \leftrightarrow X_c(j\omega) \cdot e^{-j\omega t_0}$$

$$t_0 = 2 \quad \text{rect}(t - 2) \leftrightarrow X_c(j\omega) e^{-j2\omega}$$

$$T = 2$$

$$\text{rect}\left(\frac{t - 2}{2}\right) \leftrightarrow$$

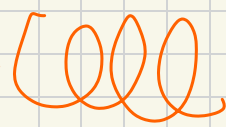
$$\text{rect}\left(\frac{t}{2}\right) \leftrightarrow 2 \cdot \text{sinc}(\omega)$$

$$\text{rect}\left(\frac{t - 2}{2}\right) \leftrightarrow 2 \cdot \text{sinc}(\omega) \cdot e^{-j2\omega}$$

$$\text{rect}\left(\frac{t}{2} - 1\right)$$

$$\frac{1}{2} \text{rect}\left(\frac{t}{2}\right)$$

$$T = 1$$



$$x(t) = e^{at} \left(\frac{t}{2} \right)$$

$$x(t-2)$$

$$\frac{d}{dt} x(t) = \frac{d}{dt} \frac{1}{2\pi} \int X(j\omega) e^{j\omega t} d\omega$$

$$\frac{d^n}{dt^n} x(t) = \frac{1}{2\pi} \int X(j\omega) (j\omega)^n e^{j\omega t} d\omega \quad (1)$$

$$\frac{d^n}{dt^n} x(t) \leftrightarrow (j\omega)^n X(j\omega)$$

$$\frac{d^n}{d\omega^n} X(j\omega) = \frac{d}{d\omega} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} x(t) (-jt)^n e^{-j\omega t} dt \quad (2)$$

$$(jt)^n x(t) \leftrightarrow \frac{d^n}{d\omega^n} X(j\omega)$$

$$x(t) = t e^{-at} u(t) \leftrightarrow X(j\omega) = ?$$

$$(-jt) e^{-at} u(t) \leftrightarrow \frac{d}{d\omega} \frac{1}{a + j\omega}$$

$$t e^{-at} u(t) \leftrightarrow j \frac{d}{d\omega} \left(\frac{1}{a + j\omega} \right)$$

$$\downarrow$$

$$j(a + j\omega) (j) (-1)$$

$$= \frac{1}{(a + j\omega)^2}$$

$$-\frac{1}{j} = j$$

$$x(at) \leftrightarrow X(j\omega)$$

$$x(at) \leftrightarrow \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau$$

$$\tau = at \quad t \rightarrow \tau/a$$

$$= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau/a} d\tau/a$$

$$= \frac{1}{a} X(j\omega/a)$$

$$a < 0$$

$$\tau = at \quad t \rightarrow \tau/a$$

$$t \rightarrow -\infty \quad \tau = +\infty$$

$$t \rightarrow \infty \quad \tau = -\infty$$

$$= \int_{\infty}^{-\infty} x(\tau) e^{-j\omega\tau/a} d\tau/a$$

$$= -\frac{1}{a} \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau/a} d\tau$$

$$= -\frac{1}{a} X(j\omega/a)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\int_{-b}^a$$

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$$x(at) \leftrightarrow \frac{1}{|a|} X(j\omega/a)$$

$$\text{rect}(3t)$$

$$\text{rect}(t) \leftrightarrow \text{sinc}(\omega/2)$$

$$\text{rect}(3t) \leftrightarrow \frac{1}{3} \text{sinc}(\frac{\omega}{6})$$

PROPERTIES OF FOURIER TRANSFORM

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$$

- Time Scaling

$$x(\alpha t) \leftrightarrow \frac{1}{|\alpha|} X\left(\frac{j\omega}{\alpha}\right)$$

where α is a real constant

- Differentiation

$$\frac{dx(t)}{dt} \leftrightarrow j\omega X(j\omega)$$

$$\frac{d^n x(t)}{dt^n} \leftrightarrow (j\omega)^n X(j\omega)$$

- Convolution

$$x(t) * h(t) \leftrightarrow X(j\omega)H(j\omega)$$

- Duality

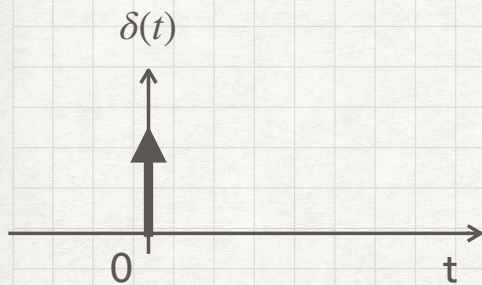
$$X(t) \leftrightarrow 2\pi x(-j\omega)$$

FOURIER TRANSFORM

IMPULSE FUNCTION



Find FT of the impulse function - $\delta(t)$



$$\text{From } X(j\omega) = \int_{t=-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$\mathcal{F}\{\delta(t)\} = \int_{t=-\infty}^{\infty} \delta(t)e^{-j\omega t} dt$$

$$= e^{-j\omega t} \Big|_{t=0}$$

$$= e^0 = 1$$

FOURIER TRANSFORM

SHIFTING



Consider the Fourier transform $X(j\omega) = 2\pi\delta(\omega - \omega_0)$.

What is the signal $x(t)$

$$\begin{aligned}x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(\omega - \omega_0) e^{j\omega t} d\omega \\&= e^{j\omega_0 t}\end{aligned}$$

$$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$$

FOURIER TRANSFORM

PERIODIC SIGNALS



Consider the periodic signal $x(t)$ with period T ; thus, $\omega_0 = 2\pi/T$. Assume $x(t)$ has a Fourier-series representation

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$$

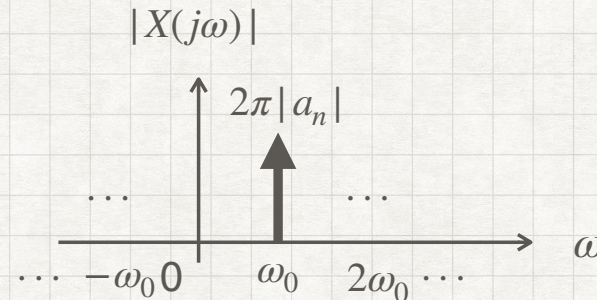
Taking the Fourier transform of both sides yields

$$X(j\omega) = \sum_{n=-\infty}^{\infty} a_n \mathcal{F}\{e^{jn\omega_0 t}\}$$

$$X(j\omega) = \sum_{n=-\infty}^{\infty} 2\pi a_n \delta(\omega - n\omega_0)$$

Fourier transform of a periodic signal is simply an impulse train with impulse located at $\omega = n\omega_0$ with strength $2\pi a_n$ and all impulses are separated from each other by ω_0 .

$x(t)$ is periodic, the magnitude spectrum $|X(j\omega)|$ is a train of impulses of strength $2\pi|a_n|$. Note that Fourier transform is not a periodic function. Even though the impulses are separate by the same amount, their weights are all different.



FOURIER TRANSFORM

IMPULSE TRAIN



Consider the periodic signal

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

To find FT, we first have to compute Fourier series coefficients

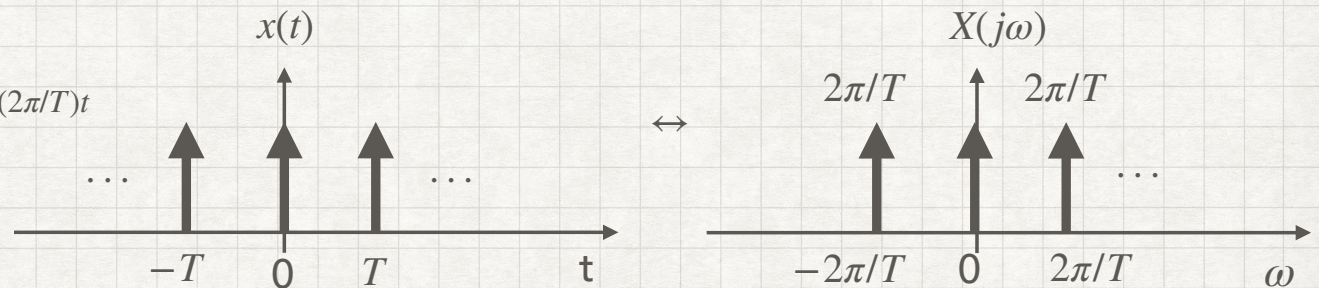
$$a_n = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jn(2\pi/T)t} dt = \frac{1}{T}$$

$$\text{From } x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(2\pi/T)t} = \sum_{n=-\infty}^{\infty} \frac{1}{T} e^{jn(2\pi/T)t}$$

Fourier transform is..

$$X(j\omega) = \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi n}{T}\right)$$

The Fourier transform of a sequence of impulses in the time domain yields a sequence of impulses in the frequency domain.



FOURIER TRANSFORM

SCALING PROPERTY

$$x(\alpha t) \leftrightarrow \frac{1}{|\alpha|} X\left(\frac{j\omega}{\alpha}\right)$$

Find $x(\alpha t)$ where α is a real constant

$$\mathcal{F}\{x(\alpha t)\} = \int_{t=-\infty}^{\infty} x(\alpha t) e^{-j\omega t} dt$$

Substitute $\tau = \alpha t$, we obtain

$$\mathcal{F}\{x(\alpha t)\} = \begin{cases} \frac{1}{\alpha} \int_{-\infty}^{\infty} x(\tau) e^{-j(\omega/\alpha)\tau} d\tau, & \alpha > 0 \\ -\frac{1}{\alpha} \int_{-\infty}^{\infty} x(\tau) e^{-j(\omega/\alpha)\tau} d\tau, & \alpha < 0 \end{cases}$$

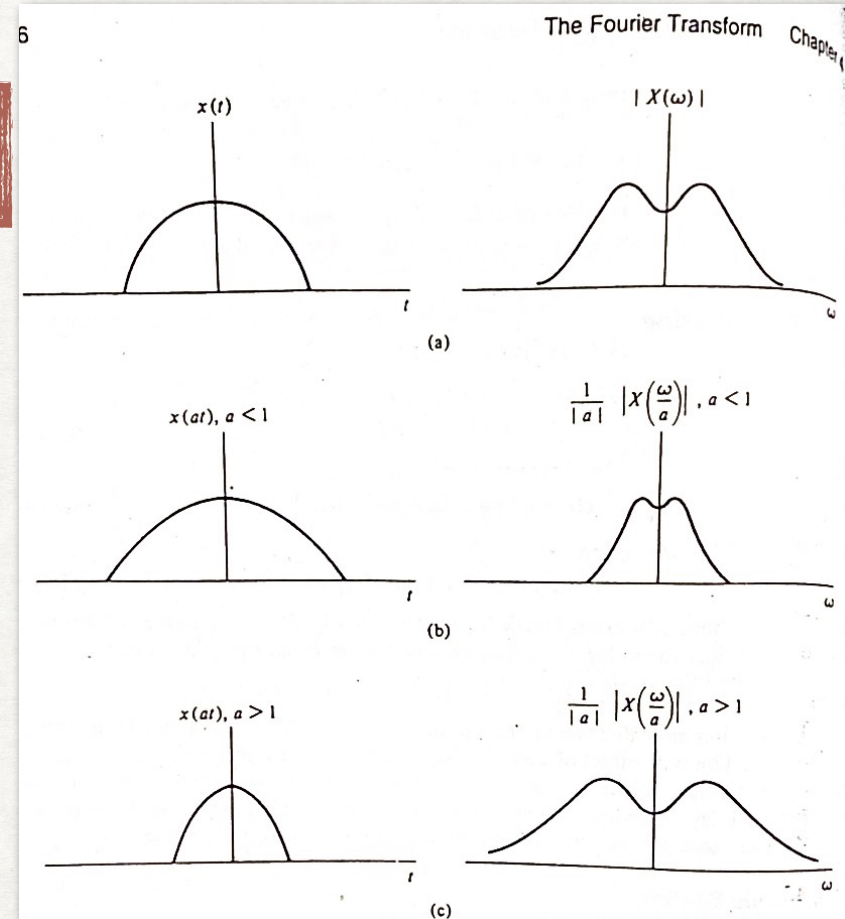


Figure 4.3.1 Examples of the time-scaling property: (a) The original signal and its magnitude spectrum, (b) the time-expanded signal and its magnitude spectrum, and (c) the time-compressed signal and the resulting magnitude spectrum.



FOURIER TRANSFORM

CONVOLUTION PROPERTY



If $x(t) \leftrightarrow X(j\omega)$, and $h(t) \leftrightarrow H(j\omega)$

Then,

$$x(t) * h(t) \leftrightarrow X(j\omega)H(j\omega)$$

The proof of this statement,...

FT

$$\mathcal{F}\{x(t) * h(t)\} = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \right] e^{-j\omega t} dt$$

conv

Interchanging the order of integration,

$$\mathcal{F}\{x(t) * h(t)\} = \int_{-\infty}^{\infty} x(\tau) \left[\int_{-\infty}^{\infty} h(t-\tau) e^{-j\omega t} dt \right] d\tau$$

From the shifting property, the bracketed term is simply $H(j\omega)e^{-j\omega\tau}$.

h(t) ↔ H(jω)
 $h(t-t_0) \leftrightarrow H(j\omega) \cdot e^{-j\omega t_0}$

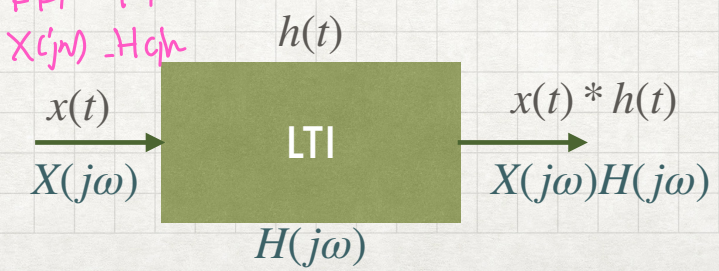
Thus,

$$\mathcal{F}\{x(t) * h(t)\} = \int_{-\infty}^{\infty} x(\tau) H(j\omega) e^{-j\omega\tau} d\tau$$

$$\mathcal{F}\{x(t) * h(t)\} = H(j\omega) \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau$$

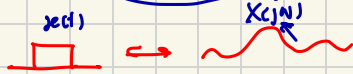
$$\mathcal{F}\{x(t) * h(t)\} = H(j\omega)X(j\omega)$$

FFT FT
 $X(j\omega) \cdot H(j\omega)$



$$\mathcal{F}\{ \text{rect}(t) * \text{rect}(t) \} \leftrightarrow \text{sinc}^2\left(\frac{\omega}{2}\right)$$

$\text{rect}(t) \leftrightarrow \text{sinc}(\omega/2)$
 $\tau = 1$



$$X_c(t) = \text{sinc}(t) \leftrightarrow x_c(\omega)?$$

$$x(t) = \frac{1}{2\pi} \int X_c(j\omega) e^{+j\omega t} d\omega$$

$$X_c(j\omega) = \int x(t) e^{-j\omega t} dt$$

$$x(\omega') = \frac{1}{2\pi} \int X_c(t') e^{+j\omega' t'} dt'$$

$$2\pi x(-\omega') = \int X_c(t') e^{-j\omega' t'} dt$$

du duality

$$X(t) \leftrightarrow 2\pi x(-\omega)$$

$$\text{rect}(-\omega) = \text{rect}(\omega)$$



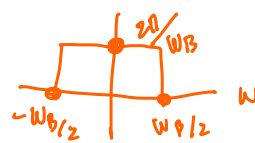
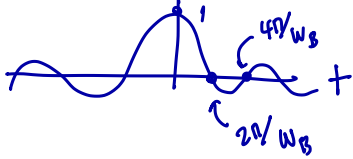
$$x(t) = \text{sinc}\left(\frac{\omega_B t}{2}\right); X_c(j\omega) = ?$$

$$\text{rect}\left(\frac{t}{\tau}\right) \leftrightarrow \tau \text{sinc}\left(\frac{\omega \tau}{2}\right)$$

$$\frac{\omega_B}{2} \text{sinc}\left(\frac{\omega_B t}{2}\right) \leftrightarrow 2\pi \text{rect}\left(\frac{\omega}{\omega_B}\right)$$

$$\text{sinc}\left(\frac{\omega_B t}{2}\right) \leftrightarrow \frac{2\pi}{\omega_B} \text{rect}\left(\frac{\omega}{\omega_B}\right)$$

$x(t)$



PROPERTIES OF FOURIER TRANSFORM

DUALITY

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

We sometimes have to find the Fourier transform of a time signal that has a form similar to an entry in the transform column in the table of Fourier transforms. We can find the desired transform by using the table backwards. To accomplish that, we write the inversion formula in the form

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \\ X(j\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \end{aligned}$$

$$\int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega = 2\pi x(t)$$

There is a symmetry between the two equations. The two equations are identical except for a sign change in the exponential, a factor of 2π and the interchange of the variables involved. This type of symmetry leads to the duality property of the Fourier transform.

Replacing t with $-t$ to get

$$2\pi x(-t) = \int_{-\infty}^{\infty} X(j\omega) e^{-j\omega t} d\omega$$

Since ω is just a dummy variable for integration, now replacing t by ω and τ by t to give..

$$X(t) \leftrightarrow 2\pi x(-j\omega)$$



FOURIER TRANSFORM

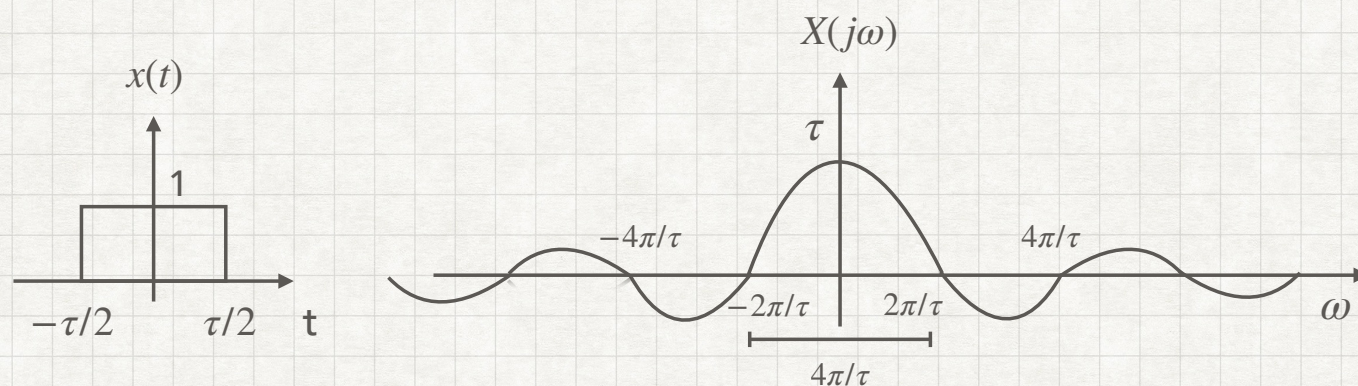
EXERCISE #3.7 DUALITY

Consider the signal

$$x(t) = \text{Sinc} \frac{\omega_B t}{2} \quad \text{Find } X(j\omega)$$

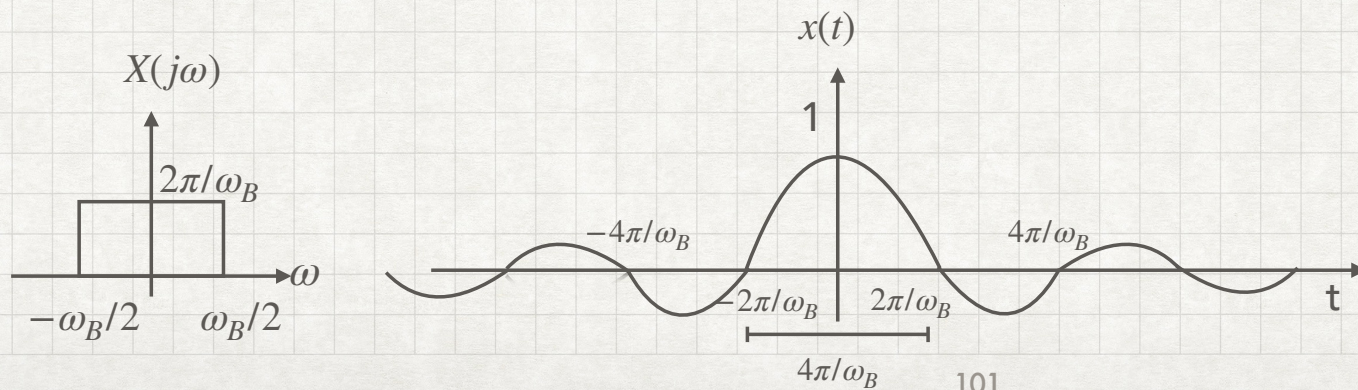
$$X(t) \leftrightarrow 2\pi x(-j\omega)$$

From Exercise #4

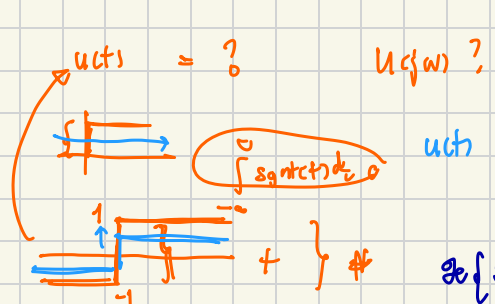


$$x(t) = \text{rect}(t/\tau)$$

$$X(j\omega) = \tau \text{sinc}(\omega\tau/2)$$



$$\delta(t) \leftrightarrow 1$$

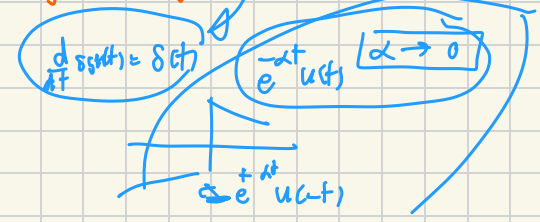


$\mathcal{L} \{1\} \leftrightarrow 2\pi \delta(\omega)$

$u(t) = \frac{1}{2} \text{sgn}(t) + \frac{1}{2}$

$= \frac{1}{2} \cdot \frac{2}{j\omega} + \frac{1}{2} \cdot 2\pi \delta(\omega) = \frac{1}{j\omega} + \pi \delta(\omega)$

signum $\rightarrow \text{sgn}(t)$



$\mathcal{F}\{\text{sgn}(t)\} = \mathcal{F}\left\{\lim_{\alpha \rightarrow 0} \left[e^{-\alpha t} u(t) - e^{\alpha t} u(-t) \right]\right\}$

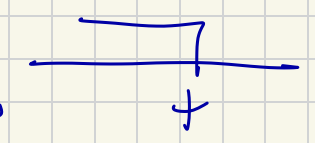
$= \lim_{\alpha \rightarrow 0} \left[\frac{1}{\alpha + j\omega} - \frac{1}{\alpha - j\omega} \right]$

$= \frac{1}{j\omega} + \frac{1}{j\omega} = \frac{2}{j\omega}$

$\frac{du(t)}{dt} = \delta(t)$

$y(t) = \int_{-\infty}^t x(\tau) d\tau$

$= x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$



$\rightarrow Y(j\omega) = ? = X(j\omega) U(j\omega) = X(j\omega) \cdot \left(\frac{1}{j\omega} + \pi \delta(\omega) \right)$

$= \frac{X(j\omega)}{j\omega} + \pi X(j\omega) \delta(\omega)$

$= \frac{X(j\omega)}{j\omega} + \pi X(j0) \delta(\omega)$

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$	$X(j\omega)$
		$y(t)$	$Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega - \theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt} x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^t x(t)dt$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \Re\{X(j\omega)\} = \Re\{X(-j\omega)\} \\ \Im\{X(j\omega)\} = -\Im\{X(-j\omega)\} \\ X(j\omega) = X(-j\omega) \\ \angle X(j\omega) = -\angle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}\{x(t)\}$ [x(t) real] $x_o(t) = \mathcal{O}\{x(t)\}$ [x(t) real]	$\Re\{X(j\omega)\}$ $j\Im\{X(j\omega)\}$

- Chapter 4 - Continuous-time Fourier transform, Signals and System, A. Oppenheim, A. Willsky and S. H. Nawab

TABLE 4.2 BASIC FOURIER TRANSFORM PAIRS

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0, \text{ otherwise}$
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0, \text{ otherwise}$
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0, \text{ otherwise}$
$x(t) = 1$	$2\pi \delta(\omega)$	$a_0 = 1, a_k = 0, k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave $x(t) = \begin{cases} 1, & t < T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t+T) = x(t)$		
	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \operatorname{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$
$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T} \text{ for all } k$
$x(t) \begin{cases} 1, & t < T_1 \\ 0, & t > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi \delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{a + j\omega}$	—
$te^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	—
$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t), \operatorname{Re}\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	—

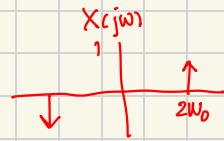
- Chapter 4 - Basic Fourier transform pairs, A. Oppenheim, A. Willsky and S. H. Nawab

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 - Chapter 3 Fourier series (3.3, 3.4)
 - Chapter 4 Fourier transform (4.2,4.3)
- Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab., **Signals and Systems**, 2nd Edition, Prentice Hall (1996)
 - Chapter 3,4

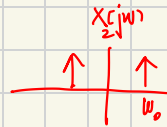
$$2) \quad x_1(t) = \sin(2\omega_0 t) = \frac{e^{j2\omega_0 t} - e^{-j2\omega_0 t}}{2j}$$

$$x_2(t) = \cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$



$$X_1(j\omega) = \frac{1}{2j} (2\pi \delta(\omega - 2\omega_0) - 2\pi \delta(\omega + 2\omega_0))$$

$$X_2(j\omega) = \frac{1}{2} (2\pi \delta(\omega - \omega_0) + 2\pi \delta(\omega + \omega_0))$$



3	$15 \cdot x(t) \leftrightarrow 15 X(j\omega)$	A
	$x(t+3) \leftrightarrow X(j\omega) \cdot e^{+j3\omega}$	D
	$x(t) \cdot e^{2jt} \leftrightarrow X(j(\omega-2))$	E
	$2 \frac{d}{dt} x(t) \leftrightarrow 2j\omega X(j\omega)$	G
	$2\pi \delta(t) \leftrightarrow 2\pi$	J